TCP sau TCP/IP

Which of the following layers is identical in both TCP/IP and OSI models?

Select one:

- a. Transport ✔
- b. Session
- c. Data Link / Link
- d. Presentation
- e. Applications

Your answer is correct.

The correct answer is: Transport

2.TCP oferă multe caracteristici avansate ce lipsesc din UDP. Care din următoarele este un avantaj a lui UDP față de TCP?

TCP provides many advanced features missing from UDP. Which of the following is an advantage that UDP has over TCP?

Select one or more:

- a. It reacts to network congestion
- b. It adds little overhead to the data transfer ✔
- c. It provides reliable data transfer
- d. It can recover gracefully from packet loss

Your answer is correct.

The correct answer is: It adds little overhead to the data transfer

3.Care din urmatoarele nu este o caracteristica a protocolului TCP?

Which of the following is not a characteristic of the TCP protocol?

Select one or more:

- a. Reliable data transfer
- b. Full-duplex operation
- c. Flow control supported
- d. Connectionless operation ✔

Your answer is correct.

The correct answer is: Connectionless operation

4.Ce înseamnă acronimul TCP?

What does the acronym TCP stand for?

Select one or more:

- a. Transmission Control Protocol ✔
- b. Transfer Correction Protocol
- c. Transition Control Protocol
- d. Transition Correction Protocol

Your answer is correct.

The correct answer is: Transmission Control Protocol

5.In timpul procesului de decapsulare in ce ordine traversam TCP/IP layers?

During the decapsulation process in what order do we traverse the TCP/IP layers?

Select one:

- a. Applications-> Internet -> Transport ->Network Access
- b. Applications-> Transport -> Network Access ->Internet
- c. Network Access-> Internet - > Transport -> Applications ✔
- d. Applications-> Transport -> Internet -> Network Access
- e. Applications-> Internet -> Network Access-> Transport

Your answer is correct.

The correct answer is: Network Access-> Internet - > Transport -> Applications



At what layer in the TCP/IP stack can we find TCP?

Select one or more:

- a. Applications
- b. Network Access
- c. Internet
- d. Transport ✔

Your answer is correct.

The correct answer is: Transport

7.Nivelul Internet din OSI este echivalent cu ce nivel din TCP/IP? (Network)

The Internet Layer from the OSI is equivalent to what layer in the TCP/IP model?

Select one:

- a. Session
- b. Transport ✘ → transport
- c. Applications
- d. Presentation
- e. Network

Your answer is incorrect.

The correct answer is: Network

8.La care nivel in TCP/IP gasim adresarea IP?

At what layer in the TCP/IP model do we find Ip addressing?

Select one or more:

- a. 2 ✔ Internet
- b. 1
- c. 4
- d. 5
- e. 3

Your answer is correct.

The correct answer is: 2

9.Care afirmatie este falsa cand vorbim despre TCP?

What statement is false when discussing TCP?

Select one or more:

- a. Ensures data delivery
- b. It has a header with a large number of bytes, at least compared to UDP ✖
- c. Does not contain in it's header a field called "Window" Liza
- d. It's header contains a checksum field

Your answer is incorrect.

The correct answer is: Does not contain in it's header a field called "Window"

10.Când închidem o sesiune TCP, ce trimite clientul?

When closing a TCP session, the client sends what?

Select one or more:

- a. The 2nd and 3rd messages
- b. The 1st and 2nd messages
- c. The 1st and 4th messages ✔
- d. The 1st and 3rd messages

Your answer is correct.

The correct answer is: The 1st and 4th messages

11.La ce nivel in modelul TCP/IP gasim DHCP?

At what layer in the TCP/IP model will we find DHCP?

Select one or more:

- a. Applications ✔
- b. Transport
- c. Internet

12.La ce nivel din stiva TCP/IP putem gasi TCP?

At what layer in the TCP/IP stack can we find TCP?

Select one or more:

- a. 2
- b. 3
- c. 4 ✖
- d. 1

Your answer is incorrect.

The correct answer is: 3

13.Nivelul de acces la rețea din TCP/IP este echivalent cu cate niveluri în modelul OSI?

The Network Access Layer from TCP/IP is equivalent to how many layers in the OSI model?

Select one:

- a. 2
- b. 1
- c. 4 ✖
- d. 3
- e. 5

Your answer is incorrect.

The correct answer is: 2

14.La ce nivel in stiva TP/IP putem gasi UDP?

At what layer in the TP/IP model stack can we find UDP?

Select one or more:

- a. 4 ✖
- b. 3
- c. 1
- d. 2

Your answer is incorrect.

The correct answer is: 3

15.La ce nivel in modelul TCP/IP gasim adresa MAC?

At what Layer in the TCP/IP model do we find MAC addresses?

Select one or more:

- a. 4
- b. 3
- c. 5
- d. 1 ✔ nivel retea
- e. 2

16.Care din urmatoarele TCP flags nu corespunde cu definitia corecta?

Which of the following TCP flags is not matched with the correct definition?

Select one or more:

- a. RST—Re-sets the sequence numbers for a TCP session
- b. ACK—Acknowledges that a TCP segment has been received
- c. FIN—Indicates the closing of a TCP session
- d. SYN—Indicates the start of a TCP connection ✖

Your answer is incorrect.

The correct answer is: RST—Re-sets the sequence numbers for a TCP session

17.La inchiderea unei sesiuni TCP, inchiderea este făcută de (message one of the begin FIN)

When discussing the closing of TCP session, that closing is made up by:

Select one:

- a. 4 messages one of them being
- b. 3 messages one of them being SYN
- c. 3 messages one of them being FIN ACK
- d. 4 messages one of them being FIN ✔

Your answer is correct.

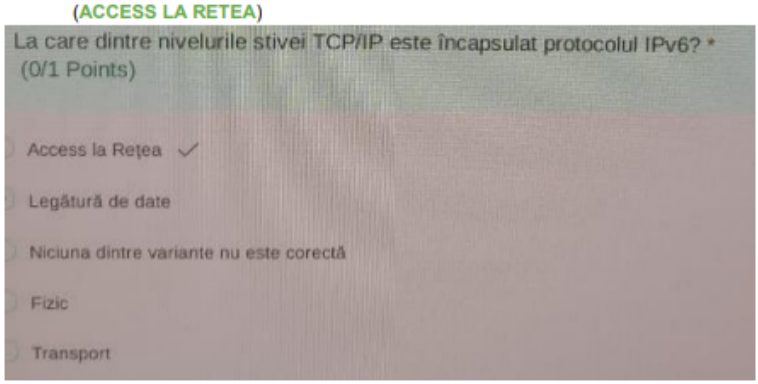
The correct answer is:
4 messages one of them being FIN

18. La care nivel din stiva TCP/IP se regaseste protocolul Ipv6? (INTERNET)

La care nivel din stiva TCP/IP se regăsește protocolul IPv6? * (1/1 Points)

- Acces la rețea
- Niciuna dintre variante nu este corectă
- Legătură de date
- Transport
- Aplicații
- Internet ✓

19. La care dintre nivelurile stivei TCP/IP este incapsulat protocolul IPV6?
(ACCESS LA RETEA)



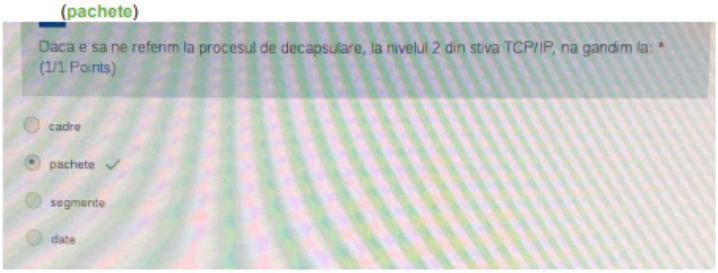
20. Dacă e sa ne referim la conceptul de cadru sau frame, le găsim în stiva TCP/IP la nivelul: (ACCES LA RETEA (NETWORK ACCESS))



21. Dacă am dori sa transmitem un email, ce protocol vom folosi la nivelul de transport? (TCP)



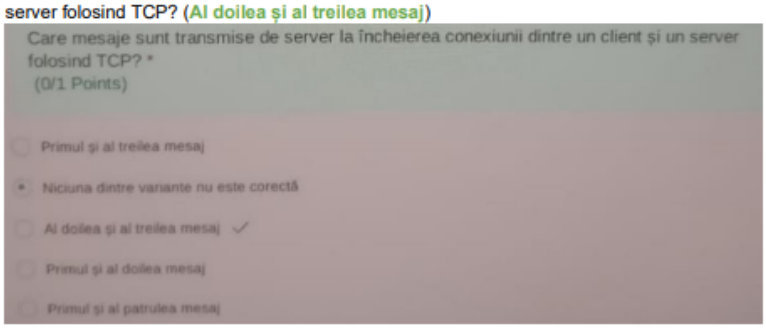
22.Dacă e sa ne referim la procesul de decapsulare, la nivelul 2 din stiva TCP/IP, ne gândim la:
(pachete)



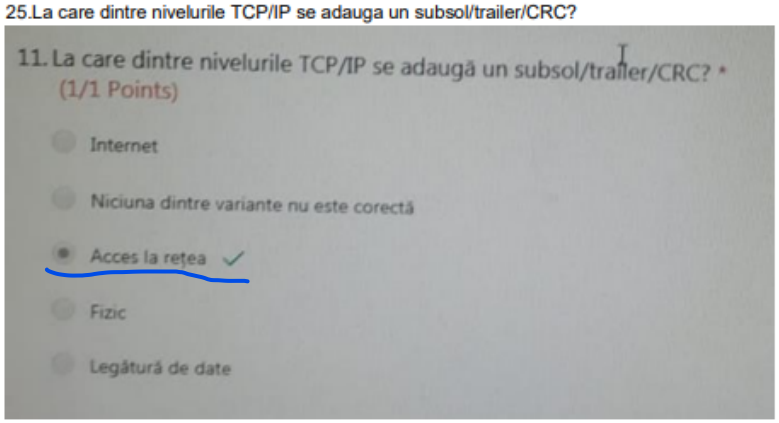
23. Dacă discutam despre three-way handshake în cadrul TCP mesajele transmise sunt:



24.Care mesaje sunt transmise de server la inchiderea conexiunii dintre un client și un server folosind TCP? (Al doilea și al treilea mesaj)

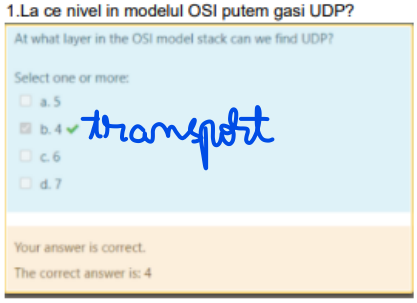


25.La care dintre nivelurile TCP/IP se adauga un subsol/trailer/CRC?



UDP

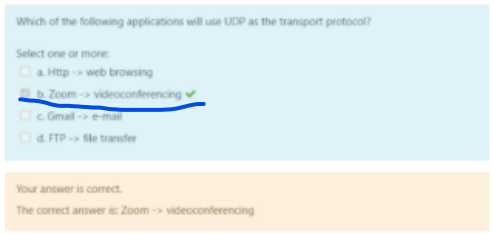
1.La ce nivel in modelul OSI putem gasi UDP?



2.Care din urmatoarele poate fi socket instantiabil cu UDP?

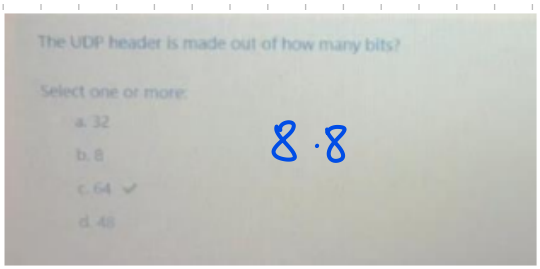


3.Care din urmatoarele aplicatii va folosi UDP ca protocol de transport?

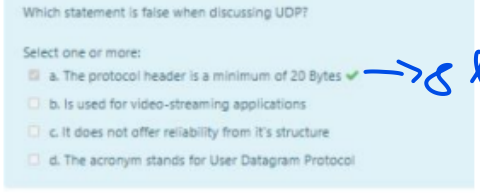


Netflix → TCP
Discord → UDP
Skype → UDP

4.Header UDP este format din cati bits?

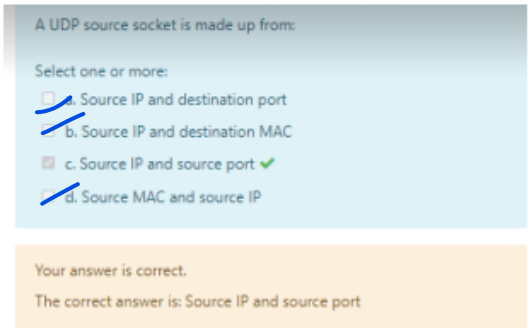


5.Care din afirmatii este falsa cand vorbim despre UDP?



→ 8 bytes maxim

6.Un socket sursa UDP este format din



7.In timpul procesului de decapsulare, la ce nivel vom elimina data specifica la UDP, la traversarea datelor conform modelului TCP/IP?

During the decapsulation process, at what layer do we remove data specific to UDP, when traversing data according to the TCP/IP model?

Select one or more:

a. 2 ✖

b. 3 → transport

c. 4

d. 1

Your answer is incorrect.

The correct answer is: 3

Activate Windows
Go to Settings to activate Wind

8.Care afirmatie este falsa cand vorbim despre UDP?

a.Offers data re-segmentation (reordering) at the reception level → dar nu oferă nici asta

b.The protocol header is composed out of 4 fields

c.It's a simple and fast protocol

d.It's basic function is reliability

9.Care din urmatoarele campuri se gaseste in antetul UDP?

Care din urmatoarele campuri se gaseste in antetul UDP? *
(1/1 Points)

Source port number ✓

sequence number doar TCP

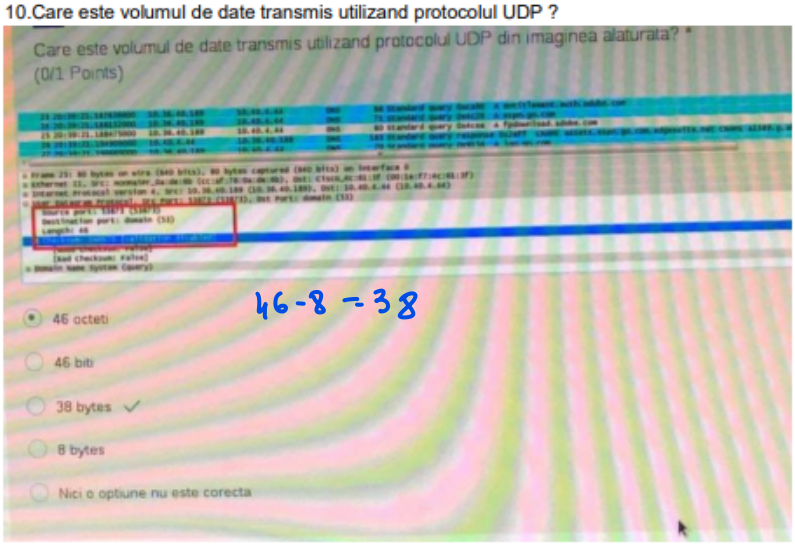
Acknowledgement number doar TCP

Control field

Window size doar TCP

10.Care este volumul de date transmis utilizand protocolul UDP ?

Care este volumul de date transmis utilizand protocolul UDP din imaginea alaturata? *
(0/1 Points)



46-8 = 38

46 octeti

46 biti

38 bytes ✓

8 bytes

Nici o optiune nu este corecta

DHCP

1.Când vom finaliza cu succes un schimb de mesaje DHCP complet, vom primi:

When successfully completing a full DHCP message exchange we will receive:

Select one or more:

- a. an IP address-> a netmask-> a gateway address + DNS
- b. an Ip address-> a gateway address-> a DNS address-> a MAC address
- c. a IP address-> a netmask-> a DNS address-> gateway address ✔
- d. a TCP address-> a gateway address -> a DNS address ->a netmask
- e. a MAC address-> a netmask-> a DNS address-> a gateway address

2.Dacă ar fi să analizăm un schimb de mesaje pentru cei mai puțini provin din DHCP, atunci al-4-lea mesaj ar fi:

If we were to look at a message exchange for the lease originate in DHCP, then the 4th message would be:

Select one or more:

- a. Acknowledge ✖ 4
- b. Request 3
- c. Discover 1
- d. No messages are delivered
- e. Offer 2

Your answer is incorrect.

The correct answer is: Request

3.Adresele primite de serverul DHCP sunt temporare?

Are the addresses received from the DHCP server temporary?

Select one or more:

- a. False
- b. True ✔
- c. Only if the server is Linux based
- d. Only if the server is Windows based

Your answer is correct.

The correct answer is: True

4.Cand ne referim la portul 67, portul folosit de obicei de către DHCP, trebuie sa afirmăm corect ca este un port din următoarea categorie:

When referring to port 67, port usually used by DHCP, we are to correctly state that it is a port from the following category:

Select one or more:

- a. Well known ports ✔ bine cunoscute 0 -> 1023

5.
Care dintre următoarele informații vor fi primite în urma apelarii cu succes a unui server DHCP?

Care dintre următoarele informații vor fi primite în urma apelării cu succes a unui server DHCP? *
(1/1 Points)

- Default Gateway, Mască de rețea, Adresă TCP, DNS
- Adresă MAC, Mască de rețea, DNS, Default Gateway
- Niciuna dintre variante nu este corectă
- Adresă IP, DNS, Mască de rețea, Default Gateway, Adresă MAC
- Mască de rețea, Default Gateway, DNS, Adresă IP ✓

MAC

1.Care afirmatie este falsa cand ne referim la adresele MAC?

Which statement is false when referring to MAC addresses?

Select one or more:

- a. They are formed from 2 components one of them identifying the producer
- b. They are formed of 48 bits
- c. They are made of 4 bytes ✔
- d. They are located on the 4th layer of the TCP/IP model stack
- e. They are unique at a global level

Your answer is correct.

The correct answer is: They are made of 4 bytes

6

2.Este adevărat ca adresele MAC se modifica în timpul unui transfer de date de la rețea la rețea?

Is it true that MAC addresses change during a data transfer from network to network?

Select one:

- True ✔
- False

The correct answer is 'True'.

3.De cate caractere hexazecimale avem nevoie pentru a avea o adresa MAC?

From how many hexadecimal characters do we need to have a MAC address?

Select one or more:

- a. 12 ✔
- b. 10
- c. 8
- d. 48
- e. 32

6.2

Your answer is correct.

The correct answer is: 12

4.Pe cati ~~biti~~ bytes reprezentam o adresele MAC?

On how many bytes do we represent MAC addresses?

Select one or more:

- a. 6 ✔
- b. 4
- c. 48
- d. 8
- e. 32

6.

Your answer is correct.

The correct answer is: 6

5.Cand un cadru cu o adresa MAC a destinației necunoscută intra într-un switch, în care port va trimite switch?

When a frame with an unknown destination MAC address enters a switch, the switch will forward it out which ports?

Select one or more:

- a. All except the port that received the frame ✔
- b. All
- c. All unicast ports
- d. None

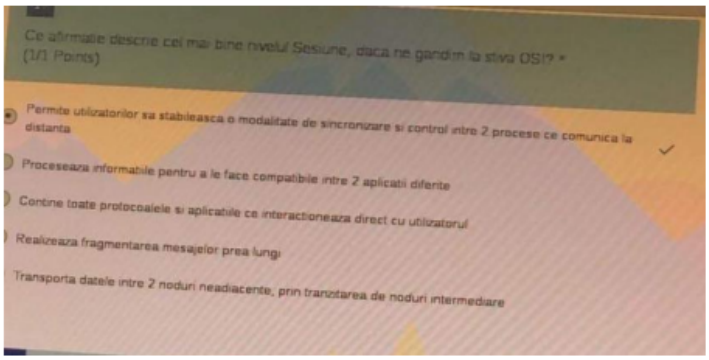
Your answer is correct.

The correct answer is: All except the port that received the frame

OSI

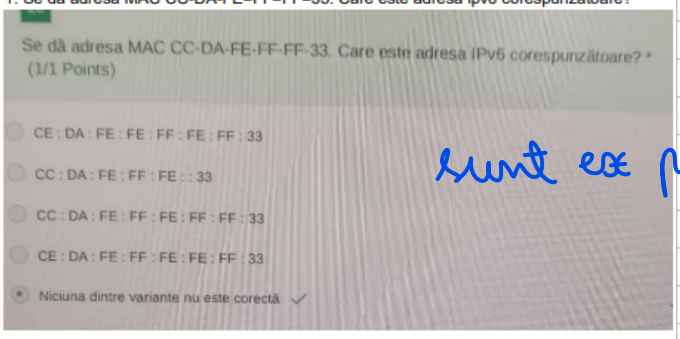
1. Ce afirmație descrie cel mai bine nivelul Sesiune, dacă ne gandim la stiva OSI?

Permite utilizatorilor să stabilească o modalitate de sincronizare și control intre 2 procese ce comunica la distanță (din laborator Gestionează sesiuni de comunicare între diverse aplicații)



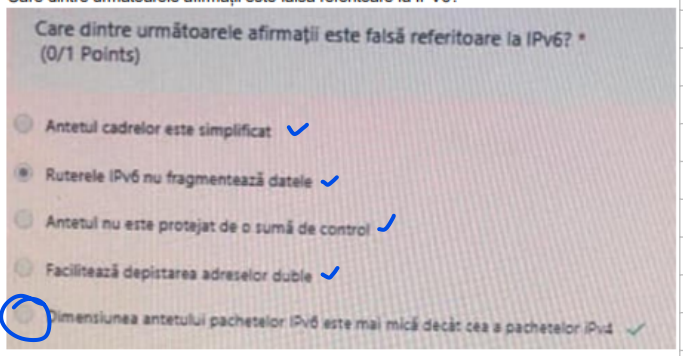
IPV6

1. Se da adresa MAC CC-DA-FE=FF=FF=33. Care este adresa ipv6 corespunzătoare?



sunt ele pe cele rezolvate

2.
Care dintre urmatoarele afirmații este falsă referitoare la IPV6?



3. Care dintre următoarele afirmații referitoare la ipv6 sunt false?
   a. Antetul pachetelor ipv6 nu este protejat de o sursa de control
   b. Nicio varianta nu e corecta
   c. Permite un numar de 10^15 adrese utile
   d. Exista o dimensiune minima a pachetelor (1280 octeti)
   e. Oferă servicii incorporate de IPsec comparativ cu ipv6

2^{128}

ADRESĂ FIZICĂ / LOGICĂ / IP

1.Care din urmatoarele este o adresa fizica valida?

Which of the following is a valid physical address?

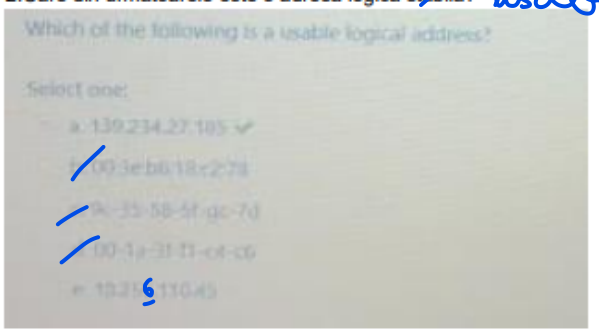
Select one:

- a. 00-1a-3f-f1-4c-c6
- b. 00:14:78:ah:9c:2b ✖
- c. 255.256.216.1
- d. 73.124.68.10
- e. 13.170.193.252

Your answer is incorrect.

The correct answer is: 00-1a-3f-f1-4c-c6

2.Care din urmatoarele este o adresa logica stabila? usable



3. Pe cati bits representam adresele fizice?

On how many bits do we represent physical addresses?

Select one or more:

- a. 6
- b. 32
- c. 48 6.8
- d. 4
- e. 8 ✖

Your answer is incorrect.

The correct answer is: 48

4.Care din urmatoarele este o adresa logica adecvata?

Which of the following is a proper logical address?

Select one:

- a. 18.256.110.45
- b. 9c-35-58-5f-4c-7d MAC
- c. 00-1a-3f-f1-c4-c6 MAC
- d. 00:3e:b6:18:c2:78 MAC
- e. 156.234.87.18 ✔

Your answer is correct.

The correct answer is: 156.234.87.18

5.Care din urmatoarele afirmatii este adevarata cand ne referim la adresele IP?

Which of the following statements is true when referring to IP addresses?

Select one or more:

- a. A physical address is unique at a LAN level
- b. A physical address is composed of 32 bits
- c. A logical address can be duplicated within a LAN nu, în LAN sunt unice
- d. A physical address is unique at a global level
- e. A logical address is unique in a LAN ✔

Your answer is correct.

The correct answer is: A logical address is unique in a LAN

6. Pe cati bytes representam o adresa IP?

On how many bytes do we represent an IP address?

Select one or more:

- [x] a. 4 ✔ IPV4
- [ ] b. 6
- [ ] c. 8
- [ ] d. 32
- [ ] e. 48

Your answer is correct.

The correct answer is: 4

7.Pe cati bits folosim pentru a reprezenta o adresa ip?

How many bits do we use to represent an ip address?

Select one or more:

- [ ] a. 48
- [ ] b. 4
- [ ] c. 6
- [ ] d. 8
- [x] e. 32 ✔ IPV4

4·8 = 32

Your answer is correct.

The correct answer is: 32

8. Care din urmatoarele afirmatii este falsa dacă ne referim la campul version din antetul IP

14

Care din urmatoarele afirmatii este falsa daca ne referim la campul version din antetul IP? *
(1 Point)

- Are ca versiune functionala IPv5 NU
- Nici o afirmatie nu este falsa
- Reprezinta versiunea protocolului ✓
- Are ca versiune functionala IPv6 ✓
- Are ca versiune functionala Ipv4 ✓

9.Care din urmatoarele adrese (fizica sau logica) poate sa apara ca și o adresa destinatie?

Care dintre următoarele adrese (fizică sau logică) poate să apară ca și o adresă destinație? *
(1/1 Points)

- FF : FE : FF : FH: FF : FF
- 135.243.215.223/27 ✓
- 110.124.26.194 255.255.254.128
- FE : 56 : AC : BG : 51 : 72
- 172.16.227.128 255.255.255.248 e dintr-o clasă privată
- Niciuna dintre variante nu este corectă

10.Care dintre urmatoarele adrese (fizica sau logica) se poate regăsi pe o interfata a unui dispozitiv?

7. Care dintre următoarele adrese (fizică sau logică) se poate regăsi pe o interfață a unui dispozitiv? *
(1/1 Points)

- 199.256.133.16 255.255.255.240
- 0.0.0.0 0.0.0.0
- AC-00-A3-B2-R3-FF
- 133.253.17.44 255.253.255.0
- Niciuna dintre variante nu este corectă ✓
- 04-D3-B0-C4-G0-72

Rutare statica / dinamica

1.Ce afirmație este corectă cand discutam despre rutare dinamica (dynamic routing)?

What statement is true when discussing dynamic routing?

Select one or more:

- a. It does not consider the addition of new nodes or links
- b. It does not deal with node or link failures
- c. Every T seconds each router sends its table to its neighbor each each router then updates its table based on the new information ✔
- d. It implies that edge costs cannot change

Your answer is correct.

The correct answer is: Every T seconds each router sends its table to its neighbor each each router then updates its table based on the new information

2.Ce afirmație este falsă atunci cand vorbim despre rutare statică? nu e automatizată

a.It implies that edge costs cannot change
b.Routers automatically transmit information from one to another in the form of routing tables
c.It does not consider the addition of new or links
d.It does not deal with node or link failures

What statement is false when discussing static routing?

Select one or more:

- a. It implies that edge costs cannot change ✔
- b. Routers automatically transmit information from one to another in the form of routing tables → e trimis dinamic
- c. It does not consider the addition of new nodes or links ✔
- d. It does not deal with node or link failures ✖ ✔

Your answer is incorrect.

The correct answer is: Routers automatically transmit information from one to another in the form of routing tables

Which of the following is not considered a routing method or protocol?

Select one or more:

- a. Open Shortest Path First (OSPF) ✔
- b. Routing Information Protocol (RIP) ✔
- c. Static routing ✖ ✔
- d. Address Resolution Protocol (ARP)

Your answer is incorrect.

The correct answer is: Address Resolution Protocol

4.
Care din următoarele procedee de rutare se folosesc cel mai eficient în rețele de dimensiuni reduse cu trafic predictibil?

5

Care din urmatoarele procedee de rutare se folosesc cel mai eficient in retele de dimensiuni reduse cu trafic predictibil? *
(1 Point)

- Rutari statice
- OSPF
- BGP
- RIP
- Nici o varianta nu este corecta

dimensiuni mari → dinamice

5. Care din urmatoarele protocoale de rutare dinamica nu este folosit pentru rutarea in interiorul unui AS ?

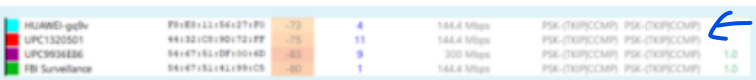
Care din urmatoarele protocoale de rutare dinamica nu este folosit pentru rutarea in interiorul unui AS *
(1 Point)

- RIP
- IGRP → ext
- OSPF
- BGP ✔ → exterior

Toate protocoalele prezentate sunt folosite pentru interior

- RIP, IGRP și OSPF sunt folosite pentru a actualiza tabelele routerelor din interioroul unui AS
- BGP este folosit pentru actualizarea tabelelor folosite de către routerele care fac legătura între AS-uri

1.În imaginea atașată este reprezentată în a 3-a coloană nivelul puterii primite în dB. Considerând doar acest parametru care este cea mai bună recepție?



-73

In the attached image, on the 3rd column is represented the received power level in dB. Considering only this parameter what is the best reception?

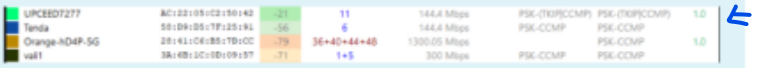
Select one or more:

- a. HUAWEI-gq9v ✔
- b. FBI Surveillance
- c. UPC1320501
- d. UPC9936E86

Your answer is correct.

The correct answer is: HUAWEI-gq9v

2.În imaginea atașată este reprezentată în a 3-a coloană nivelul puterii primite în dB. Considerând doar acest parametru care este cea mai bună recepție?



-21

In the attached image, on the 3rd column is represented the received power level in dB. Considering only this parameter what is the best reception?

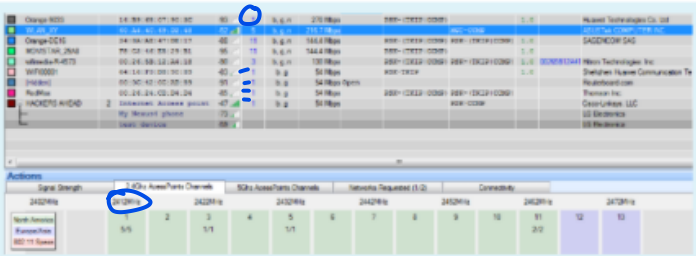
Select one or more:

- a. vali1
- b. Tenda
- c. UPCED7277
- d. Orange-hD4P-5G ✖

Your answer is incorrect.

The correct answer is: UPCED7277

3.Cate rețetele sunt suprapuse pe canalul cu frecvența centrală 2412MHz?



How many networks do we have overlapping on the channel with the central frequency 2412MHz?

Select one or more:

- a. 3
- b. 4
- c. 1
- d. 2
- e. 5 ✔

Your answer is correct.

The correct answer is: 5

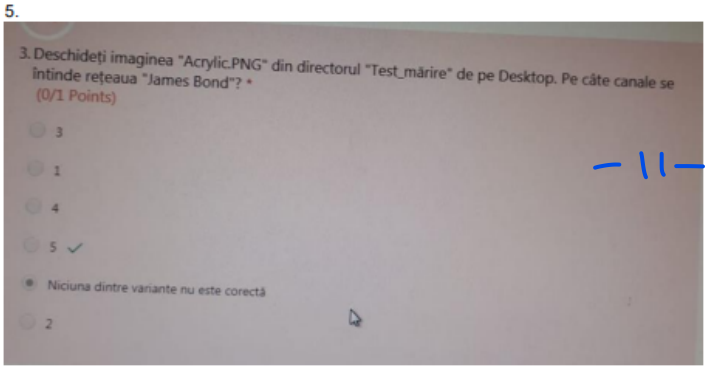
4. Tinand cont de imaginea "Acrylic_wifi", obtinuta cu instrumentul Acrylic Wifi Home, ce afirmație este corectă:

Tinand cont de imaginea "Acrylic_wifi", obtinuta cu intrumentul Acrylic WiFi Home, ce afirmatie este corecta : *
(1 Point)

- Reteaua cu SSID-ul Cloud se va receptiona mai bine decat cea cu SSID-ul ACSAir
- Reteaua transmisa folosind 2 canale radio este UPT-eduroam
- Reteaua cu SSID-ul LRG poate avea un debit pe spectrul radio de 300Mbps
- Reteaua cu SSID-ul B414 are o putere de transmisie de -64dB

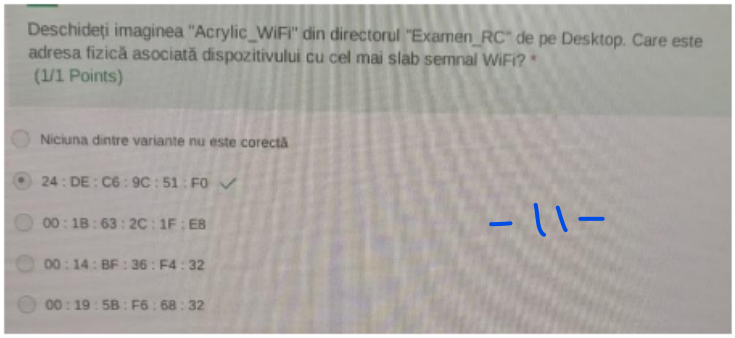
nu avem imagine

5.



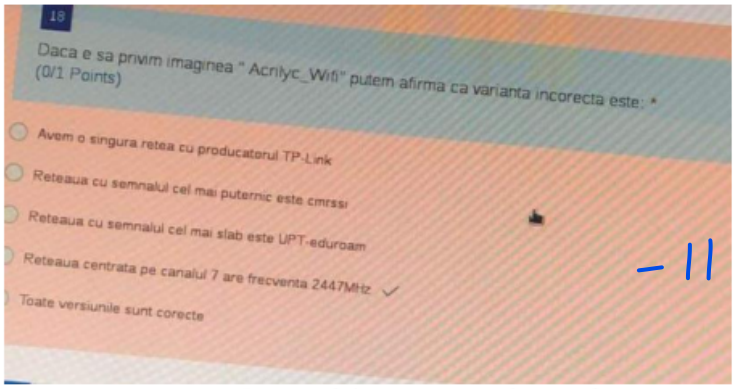
- 11 -

6.



- 11 -

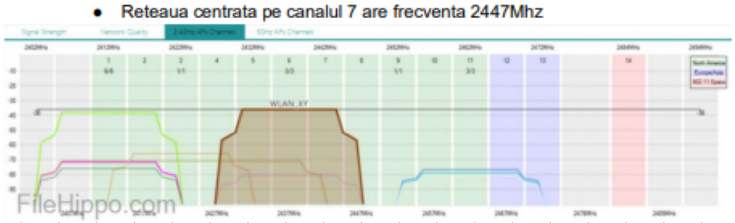
7.



- 11 -

8. Daca e sa privim imaginea "Acrylic_Wifi" (nu stiu unde e poza): putem afirma ca varianta incorecta este:

- Reteaua centrata pe canalul 7 are frecventa 2447Mhz



ARP

1.Protocolul ARP face legatura intre:

Adresa MAC si o adresa ip

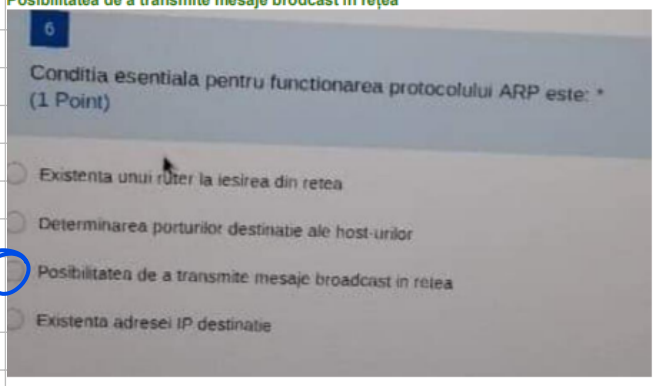
18

Protocolul ARP face legatura intre *
(1 Point)

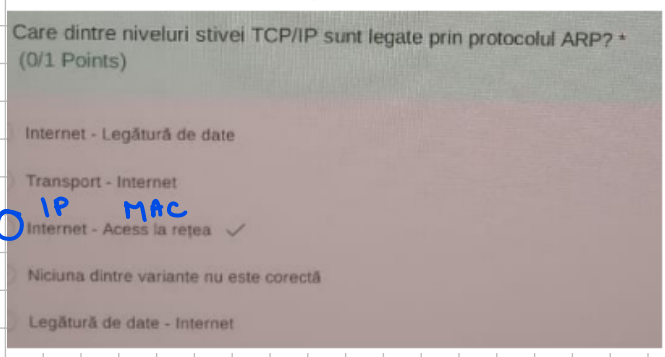
- Adresa MAC si o adresa IP
- adresa fizica si un port sursa
- Nici o varianta nu este corecta
- adresa logica si portul destinatie
- adresa MAC si o adresa fizica

2.Conditia esentiala pentru functionarea protocolului ARP este :

Posibilitatea de a transmite mesaje brodcast in rețea



3.Care dintre nivelurile stivei TCP/IP sunt legate prin protocolul ARP?



Comenzi

1.Care comanda windows ne-ar oferi output-ul?

```
Internet Address      Physical Address      Type
135.243.230.1         2c-fa-a2-49-b2-ce     dynamic
135.243.230.47        f8-75-a4-13-98-1c     dynamic
135.243.230.62        98-fa-9b-12-16-70     dynamic
135.243.230.69        3c-18-a0-b2-b1-88     dynamic
135.243.230.185       8c-16-45-ea-90-31     dynamic
135.243.230.195       8c-16-45-62-ed-e4     static
135.243.231.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Which windows command would give us the attached output?

Select one:

- a. ipconfig
- b. arp -a
- c. show internet address
- d. netstat ✖
- e. ipconfig /all

Your answer is incorrect.

The correct answer is: arp -a

2.Care comanda windows ne-ar oferi output-ul? (Daca era ping -> era bine)

```
Reply from 10.5.55.248: bytes=32 time=3ms TTL=57
Reply from 10.5.55.248: bytes=32 time=4ms TTL=57
Reply from 10.5.55.248: bytes=32 time=6ms TTL=57
Reply from 10.5.55.248: bytes=32 time=3ms TTL=57

Ping statistics for 10.5.55.248:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 4ms
```

Which windows command would give us the attached output?

Select one:

- a. ipconfig
- b. arp -a
- c. traceroute ✖
- d. No command in this form would give this output
- e. neststat

Your answer is incorrect.

The correct answer is: No command in this form would give this output

3.Care comanda windows ne-ar oferi output-ul?

```
Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::c055:6a23:4f28:4dc3%17
Autoconfiguration IPv4 Address. . : 169.254.77.195
Subnet Mask . . . . . . . . . . . : 255.255.0.0
Default Gateway . . . . . . . . . :
```

Which windows command would give us the attached output?

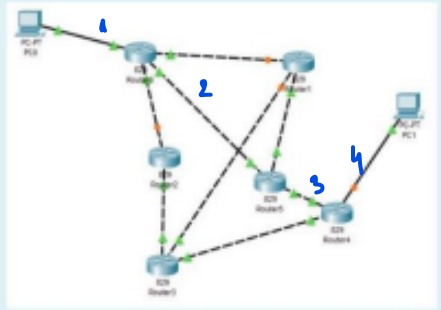
Select one or more:

- a. arp -a
- b. traceroute
- c. ipconfig ✔
- d. ipconfig /all
- e. ping

Your answer is correct.

The correct answer is: ipconfig

1.Care este valoarea minima TTL pentru avea o conexiune între 2 PC-uri. Forma liniilor, și culoarea punctelor și triunghiurilor nu trebuie sa fie luat în considerare.



What is the minimum TTL value in order to have a connection between the 2 PC's. The shape of the lines, and colour of the bullets and triangles is not to be taken into account.

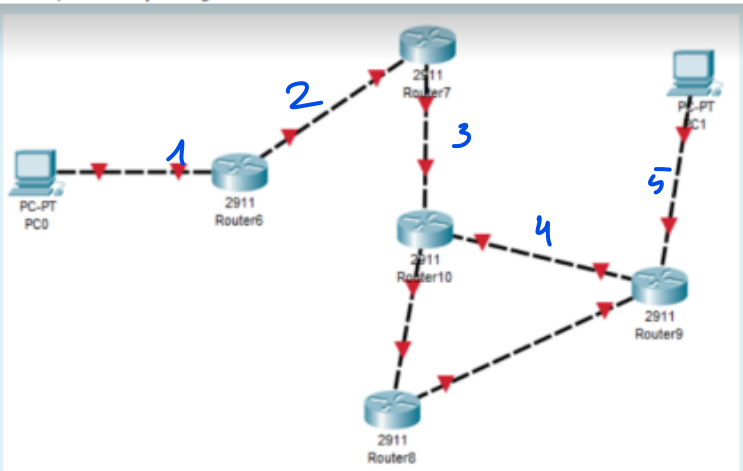
Select one:

- a. 3 ✖
- b. We will not have connection regardless of the TTL value
- c. 5
- d. 6
- e. 4

Your answer is incorrect.

The correct answer is: 4

2.Care este valoarea minima TTL pentru avea o conexiune între 2 PC-uri. Forma liniilor, și culoarea punctelor și triunghiurilor nu trebuie sa fie luat în considerare.



What is the minimum TTL value in order to have a connection between the 2 PC's. The shape of the lines, and colour of the bullets and triangles is not to be taken into account.

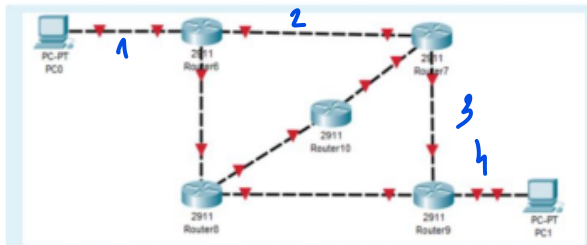
Select one or more:

- a. We will not have connection regardless of the TTL value ✖
- b. 3
- c. 4
- d. 6
- e. 5

Your answer is incorrect.

The correct answer is: 5

3.Care este valoarea minima TTL pentru avea o conexiune între 2 PC-uri. Forma liniilor, și culoarea punctelor și triunghiurilor nu trebuie sa fie luat în considerare.



What is the minimum TTL value in order to have a connection between the 2 PC's. The shape of the lines, and colour of the bullets and triangles is not to be taken into account.

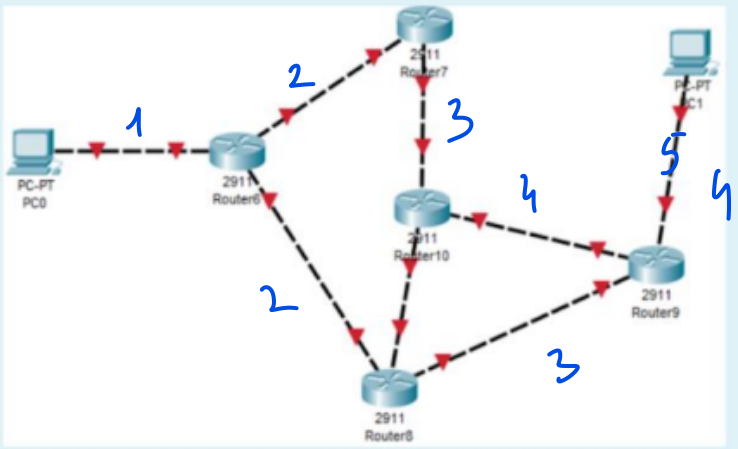
Select one or more:

- a. 2
- b. 4
- c. We will not have connection regardless of the TTL value ✖
- d. 5
- e. 3

Your answer is incorrect.

The correct answer is: 4

4.Care este valoarea minima TTL pentru avea o conexiune între 2 PC-uri. Forma liniilor, și culoarea punctelor și triunghiurilor nu trebuie sa fie luat în considerare.

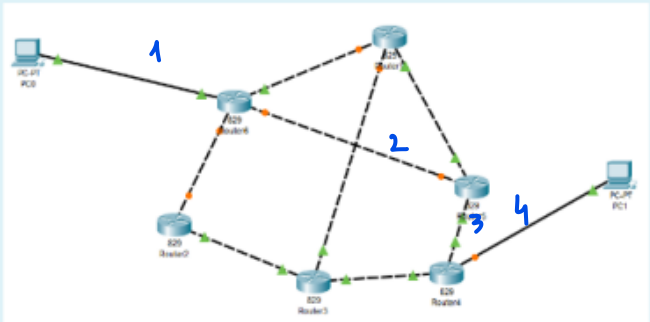


What is the minimum TTL value in order to have a connection between the 2 PC's. The shape of the lines, and colour of the bullets and triangles is not to be taken into account.

Select one or more:

- a. 6
- b. We will not have connection regardless of the TTL value
- c. 5
- d. 4 ✔
- e. 3

5.Care este valoarea minima TTL pentru avea o conexiune între 2 PC-uri. Forma liniilor, și culoarea punctelor și triunghiurilor nu trebuie sa fie luat în considerare.



What is the minimum TTL value in order to have a connection between the 2 PC's. The shape of the lines, and colour of the bullets and triangles is not to be taken into account.

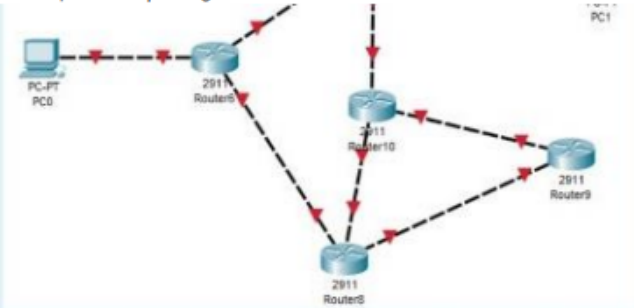
Select one:

- a. 4 ✔
- b. We will not have connection regardless of the TTL value
- c. 3
- d. 5
- e. 6

Your answer is correct.

The correct answer is: 4

6.Care este valoarea minima TTL pentru avea o conexiune între 2 PC-uri. Forma liniilor, și culoarea punctelor și triunghiurilor nu trebuie sa fie luat în considerare.

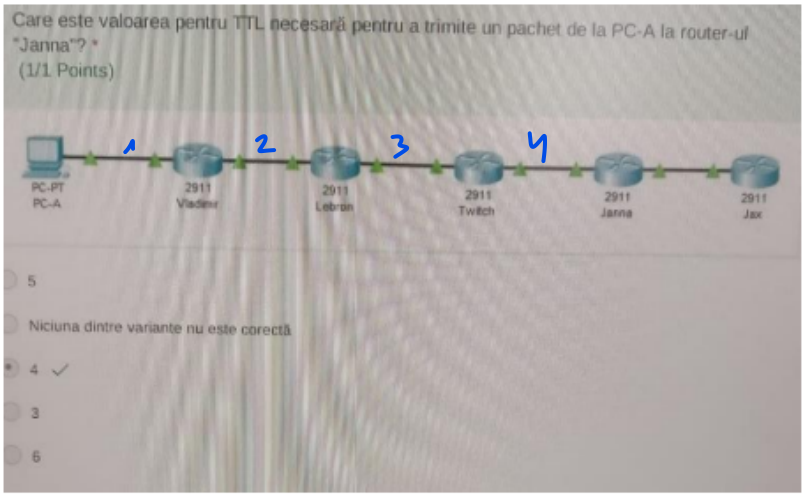


What is the minimum TTL value in order to have a connection between the 2 PC's. The shape of the lines, and colour of the bullets and triangles is not to be taken into account.

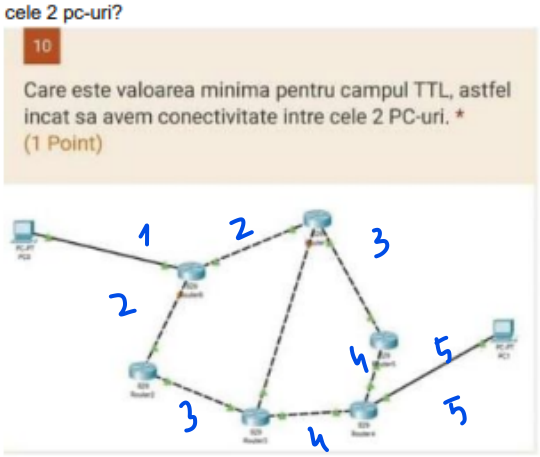
Select one or more:

- a. 6
- b. 3
- c. 5
- d. We will not have connection regardless of the TTL value ✔
- e. 4

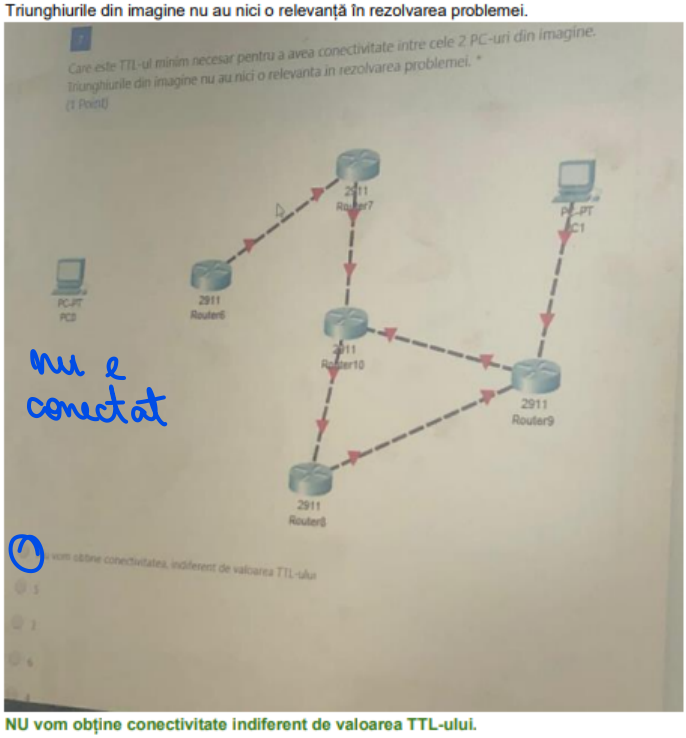
7.Care este valoare pentru TTL necesara pentru a trimite un pachet de la PC-A la router-ul "Janna"?



8.Care este valoarea minima pentru campul TTL, astfel incat sa avem conectivitate între cele 2 pc-uri?



9.Care este TTL-UL minim necesar pentru a a cea conectivitate intr 2 PC-uri din imagine. Triunghiurile din imagine nu au nici o relevanță în rezolvarea problemei.



NU vom obține conectivitate indiferent de valoarea TTL-ului.

Broadcast

1.Care din urmatoarele afirmatii este gresit cand vorbim despre broadcast?

What of the following statements is wrong when discussing broadcast?

Select one or more:

- a. At the Network Access layer the broadcast address is FF-FF-FF-FF-FF-FF
- b. You need the subnet mask to compute the broadcast address in logical addressing
- c. You can apply the concept only to the Transport Layer
- d. We discuss one emitter and all receivers from a single network
- e. It is not dependent on the network topology used

merge și la altele

Your answer is incorrect.

The correct answer is: You can apply the concept only to the Transport Layer

2.Dacă discutam despre o comunicare de tip broadcast, care dintre urmatoarele afirmatii descrie cel mai bine acest tip de comunicare

Este o comunicatie ce are ca receptori toate hosturile din rețea

13

Daca discutam despre o comunicatie de tip broadcast, care dintre urmatoarele afirmatii descrie cel mai bine acest tip de comunicatie : *
(1 Point)

- Este o comunicatie ce are ca receptori toate host-urile din retea
- Este o comunicatie in care sursa transmite si un singur receptor asculta
- Nici o varianta nu este corecta
- Este o comunicatie in care avem mai multe surse si mai multi receptori
- Este formata dintr-o sursa si mai multi dar nu toti receptorii

multicast

Simplex, fullduplex/ multicast/full mesh/unicast

1.Dacă urmează sa primim o pagina de la un scaner al unui birou de imprimante , ce tip de comunicare ar trebuie sa folosim?

If we are going to receive a page from the scanner of a centralised office printer, what type of communication would be used?

Select one or more:

- a. Broadcast
- b. Simplex ✔
- c. Autonomous
- d. Full-duplex
- e. Multicast

Your answer is correct.

The correct answer is: Simplex

2. Dacă participam într-o conferința Zoom care încurajează și permite participarea, ce tip de comunicare ar descrie cel mai bine aceasta situatie?

If we are to participate in a Zoom conference that encourages and allows participation, which type of communication would best describe this situation?

Select one or more:

- a. Full-duplex ✔
- b. Full-mesh
- c. Simplex
- d. Partial-mesh
- e. Half-duplex

Your answer is correct.

The correct answer is: Full-duplex

3.Ce descriere este cea mai buna pentru full-mesh topology?

What description best fits the full-mesh topology?

Select one or more:

- a. One element transmits and all other receive
- b. Any element can transmit only when it receives a token
- c. All elements transmit at the same time
- d. Every element is connected to all others ✔
- e. The transmition is done sequentially

Your answer is correct.

The correct answer is: Every element is connected to all others

4.Care din urmatoarele tehnologii asigura ca un pachet unicast este vizibil numai pentru device-ul cu o specifică adresa destinatie?

Which of the following technologies ensures that a unicast packet is visible only to the device with the specific destination address?

Select one or more:

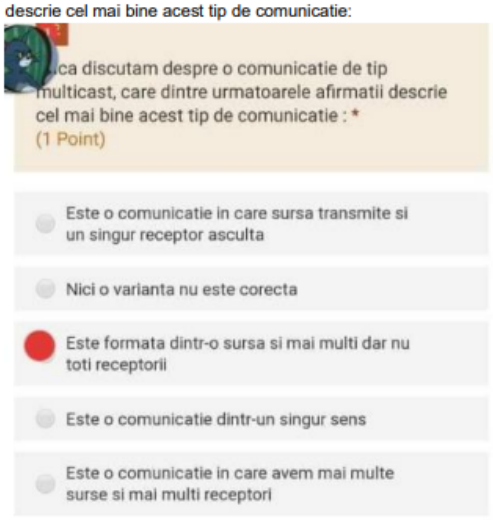
- a. Satellite
- b. Ethernet
- c. Wireless LAN
- d. Switched Ethernet ✔

Ethernet e prin cablu

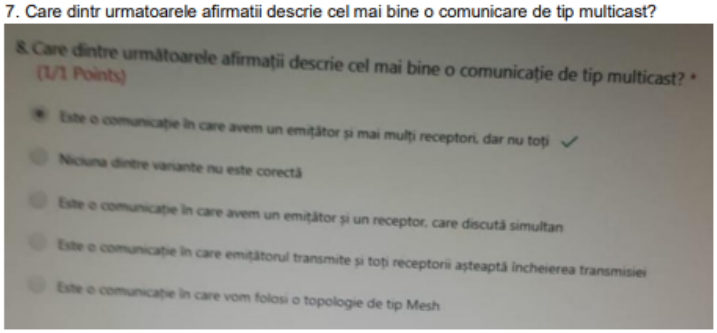
Your answer is correct.

The correct answer is: Switched Ethernet

5. Dacă discutam despre o comunicare de tip multicast, care dintre urmatoarele afirmatii descrie cel mai bine acest tip de comunicatie:

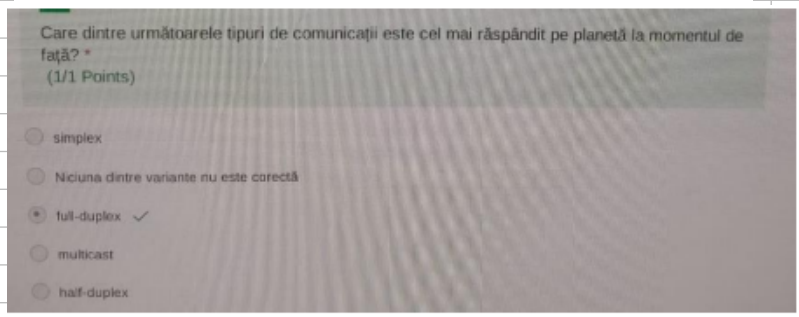


7. Care dintr urmatoarele afirmatii descrie cel mai bine o comunicare de tip multicast?



A. Este formata dintr-o sursa si mai multi dar nu toti receptorii
B. Este o comunicatie in care avem un emitator si mai multi receptori dar nu toti.

6. Care dintre următoarele tipuri de comunicatii este cel mai raspandit pe planetă?



Ethernet / wireless/ router

1.Cand discutam despre mediul wireless, ce afirmație este falsă?

When discussing the wireless medium, what is the false statement?

Select one or more:

- a. We can use light pulses
- b. We can use infrared ✗ ✓ → nu e fiabilă
- c. We can use radio waves ✓
- d. We can use microwaves ✓

Your answer is incorrect.

The correct answer is: We can use light pulses

2. Ce afirmație ce se referă la Ethernet este falsă?

Which statement regarding Ethernet is false?

Select one or more:

- a. It uses transceivers at the ends, equipment that can send and receive data ✓
- b. It is based on Link state algorithms → e la rutare
- c. It uses CSMA/CD ✗ ✓
- d. Recently the standard included technologies that allow data transfers up to 1Gbps ✓

Your answer is incorrect.

The correct answer is: It is based on Link state algorithms

3.Evaluați urmatoarea afirmatie "Un router conecteaza echipamente din 2 rețele diferite"

Evaluate the following statement. "A router connects equipment from 2 different networks. "

Select one:

- True ✓
- False

The correct answer is 'True'.

4.Care din urmatoarele tehnologii asigura ca un pachet unicast este vizibil numai pentru device-ul cu o specifică adresa destinatie?

Which of the following technologies ensures that a unicast packet is visible only to the device with the specific destination address?

Select one or more:

- a. Switched Ethernet ✓
- b. Wireless LAN
- c. Ethernet
- d. Satellite

Your answer is correct.

The correct answer is: Switched Ethernet

5.Care din urmatoarele nu e o tipologie de retea?

Which of the following is not a network topology?

Select one or more:

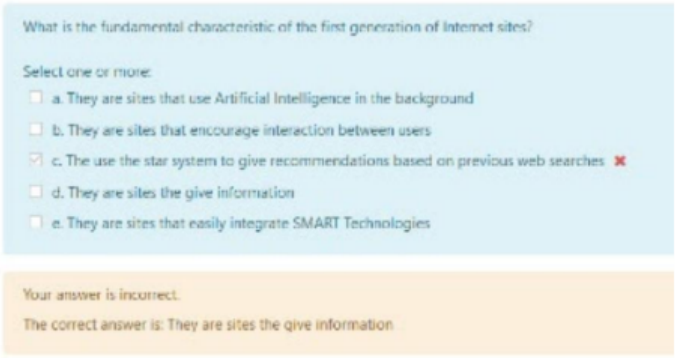
- a. Autonomous ✓
- b. Poit-to-point
- c. Token Ring
- d. Star
- e. Mesh

Your answer is correct.

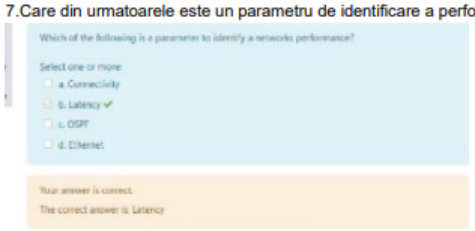
The correct answer is: Autonomous

6.Ce este caracteristică fundamentală a primei generatii de site-uri internet.

a .They are sites that use Artificial Intelligence in the background
b. They are sites that encourage interaction between users
c. The use the star system to give recommendations based on previous web searches
d. They are sites the give information
e. They are sites that easily integrate SMART Technologies

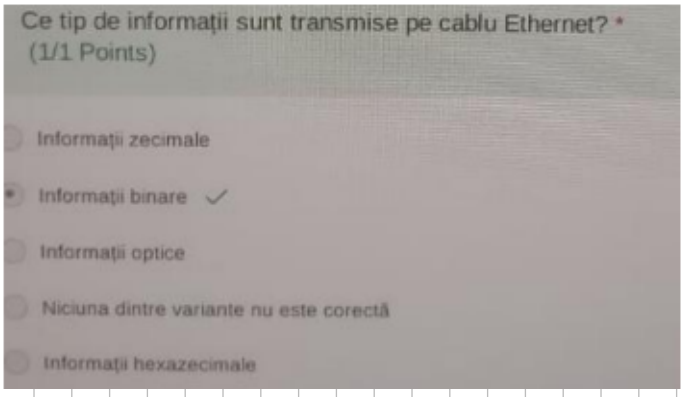


7.Care din urmatoarele este un parametru de identificare a performanței rețelei? -> Latența



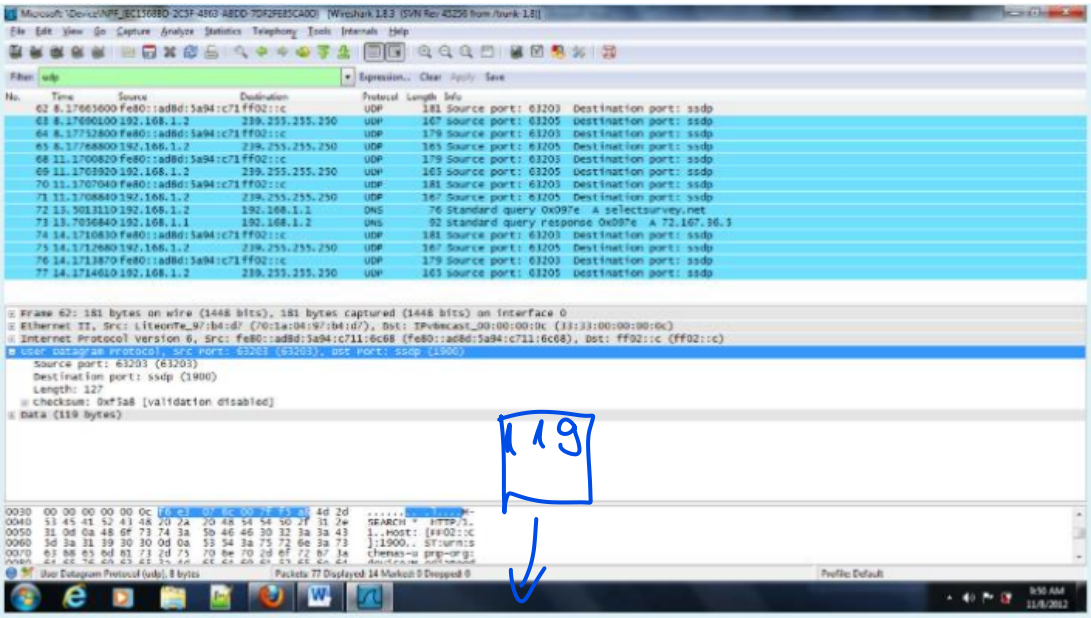
8. Ce tip de informații sunt transmise pe cablu Ethernet?

INFORMATII BINARE



Wireshark

1.



Using the provided image, compute the value of the payload in bytes

Select one:

a. 120

b. None of the provided values is the correct one

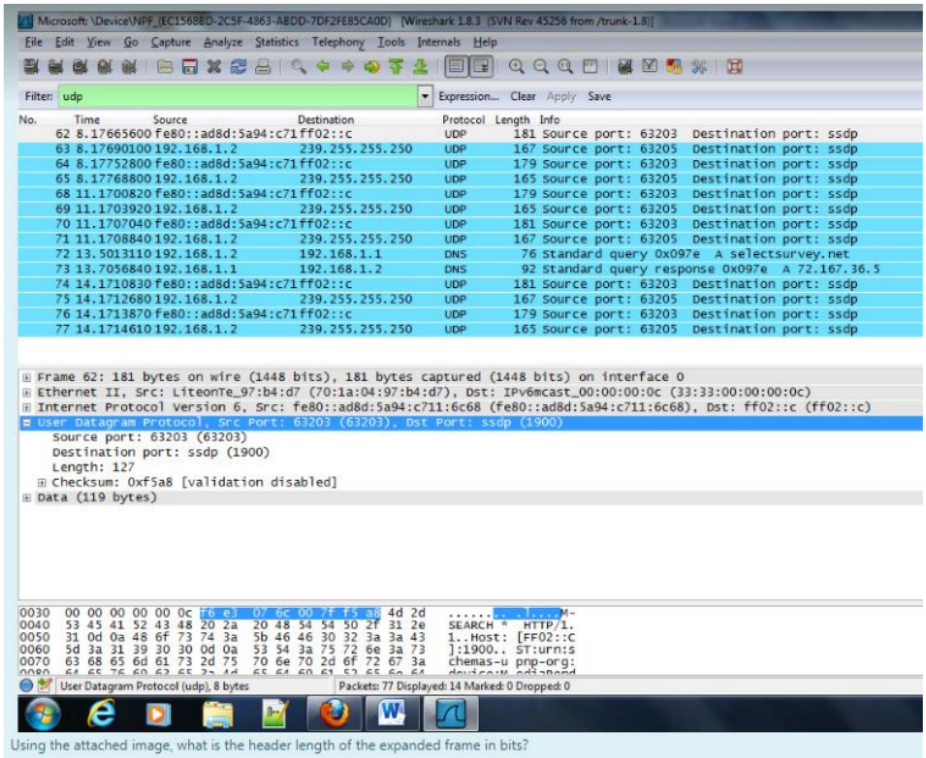
c. 127 ✖

d. 118

Your answer is incorrect.

The correct answer is: None of the provided values is the correct one

2.



Using the attached image, what is the header length of the expanded frame in bits?

Select one:

a. 127

b. 64 ✔

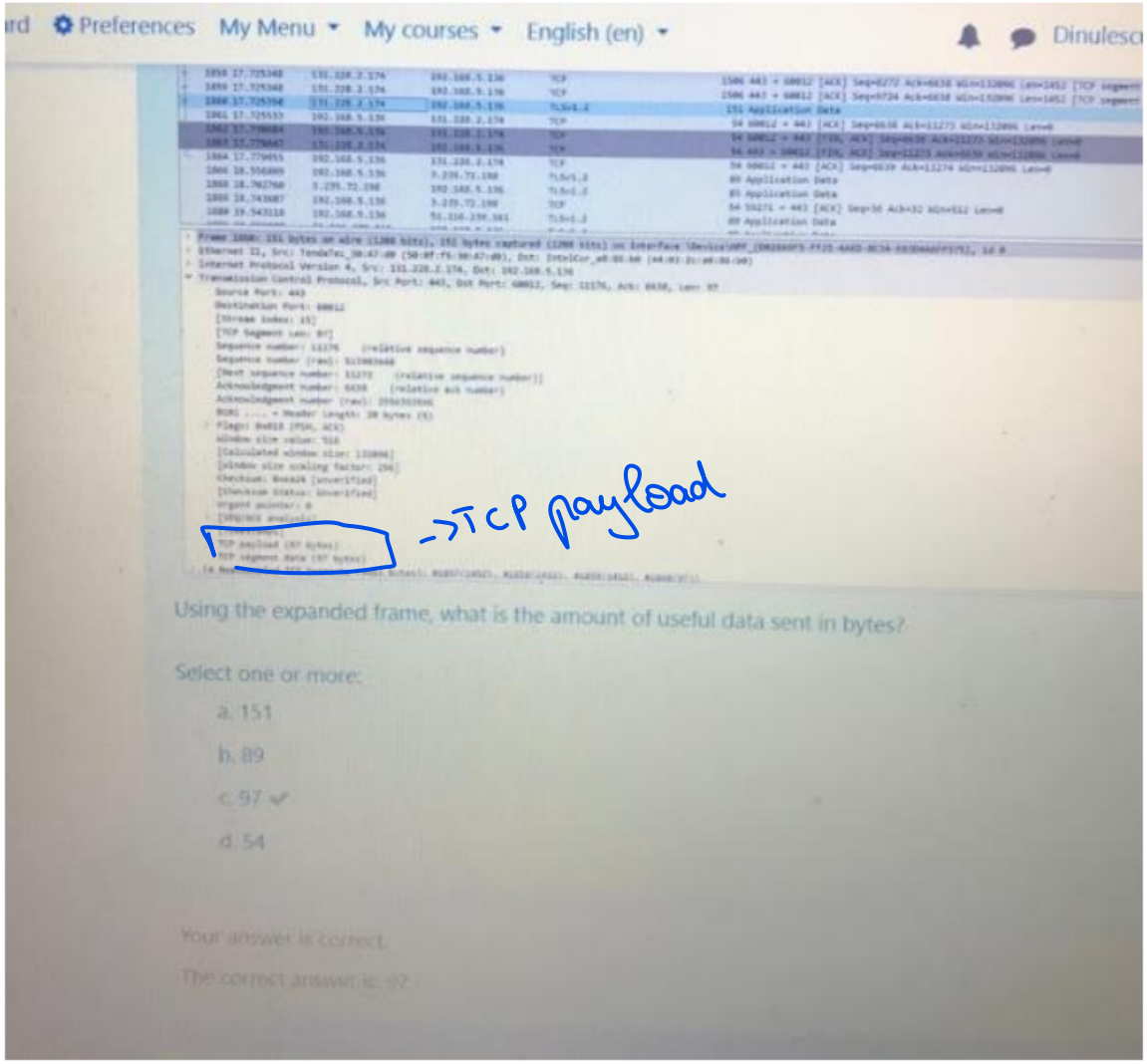
c. 8

d. 1016

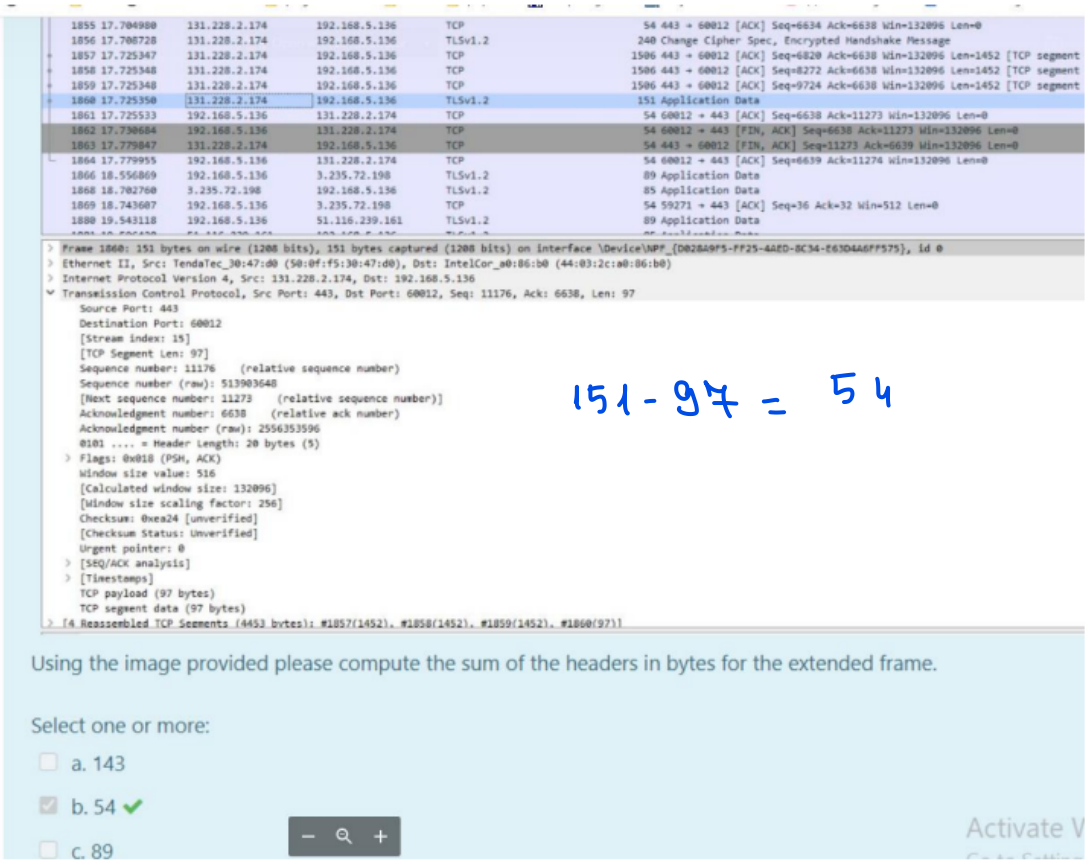
UDP -> mehren 8 bytes

=> 8·8 = 64

3.



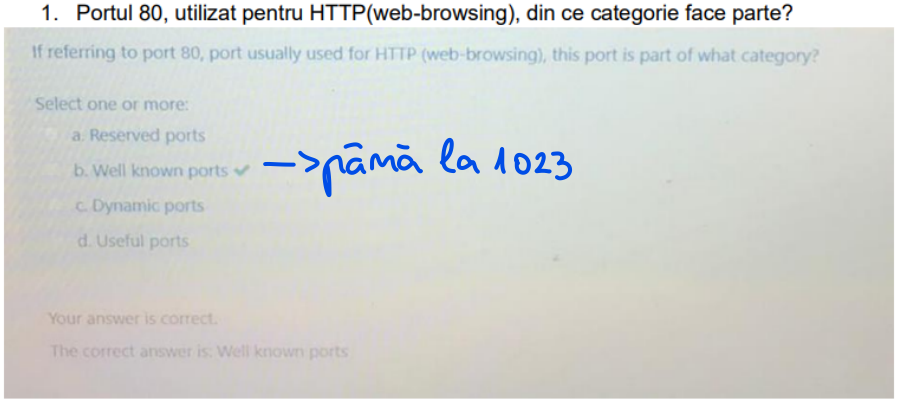
4.



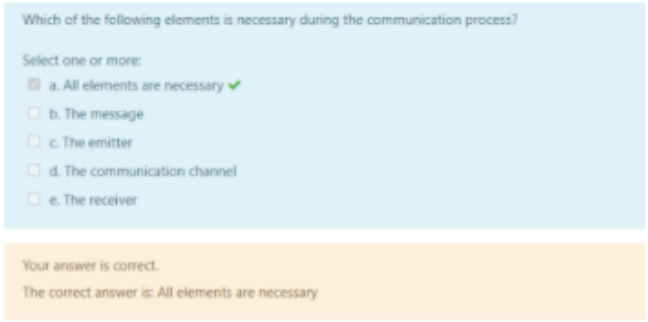
- 54 (lungimea cadrului-lungimea datelor = 151-97)

Portul 80 / comunicatii

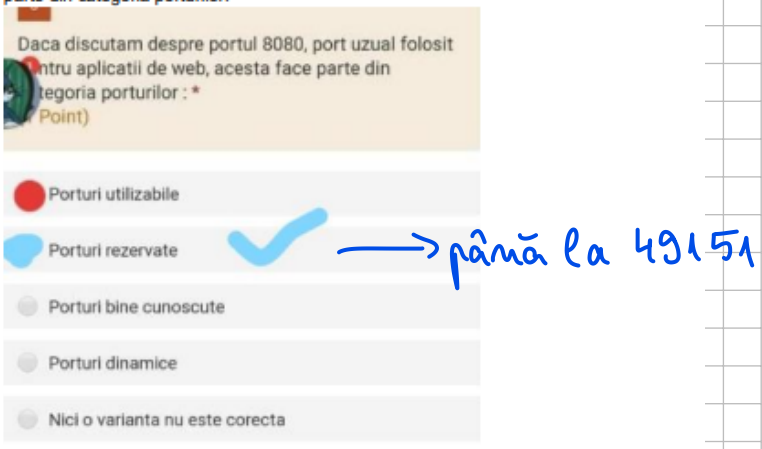
1. Portul 80, utilizat pentru HTTP(web-browsing), din ce categorie face parte?



2.Care din urmatoarele elemente este necesar in timpul procesului de comunicare?



3. Dacă discutam despre portul 8080, port uzual folosit pentru aplicații web, acesta face parte din categoria porturilor:



4.Pentru aplicațiile de e-mail exita disponibile diverse protocoale, unul dintr ele fiind POP3. Aceasta folosește ca și destinație portul 110. Acest port face parte din categoria porturilor



Ping / tracert / hubs

1.Daca ne referim la comand ping putem spune ca:

If we refer to the ping command we can say that:

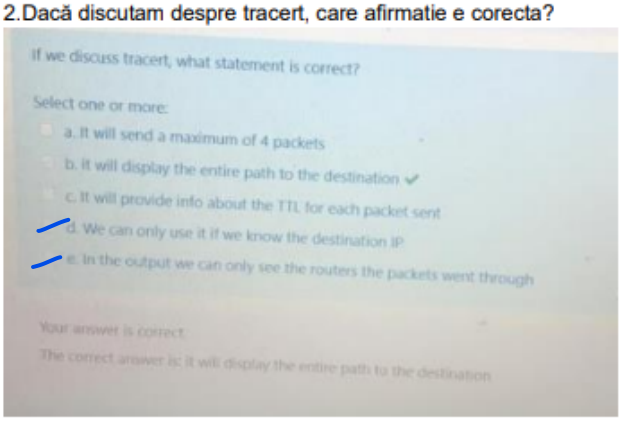
Select one:

- a. Can return the message "request time out"
- b. Shows us the time needed for the pachet to get to the destination and back again
- c. On windows it will send 4 packets each of 32 bytes
- d. All variants are correct ✔
- e. Its purpose is to check the connection to a distant device

Your answer is correct.

The correct answer is: All variants are correct

2.Dacă discutam despre tracert, care afirmatie e corecta?



3.Care din urmatoarele afirmatii este falsa?

Which of the following statements is false?

Select one or more:

- a. Hubs provide collision domain separation ✔
- b. Switches provide collision domain separation ✔
- c. VLANs provide broadcast domain separation ✔
- d. Which of the following statements is false?

Your answer is correct.

The correct answer is: Hubs provide collision domain separation

4.Daca ne referim la comanda tracert, ce afirmație este corectă?

If we refer to the tracert command, what is the correct statement?

Select one or more:

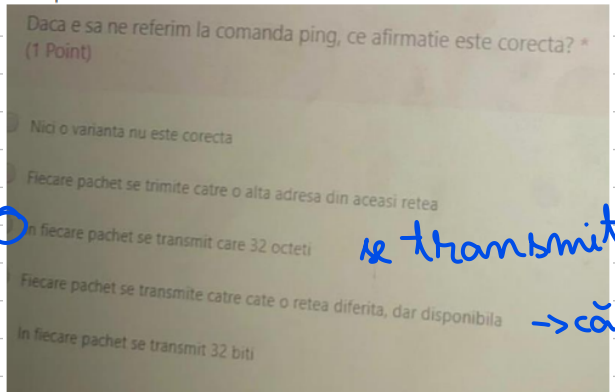
- a. In order to reach the destination we must not exceed the TTL value — nu
- b. It will show us only the time to reach the destination ✖ — nu + path
- c. It will send a maximum of 4 packets of 32 bytes — → ping
- d. We can only use it if we know the destination IP — nu
- e. It will check the availability of a distant host without giving any other output — nu

Your answer is incorrect.

The correct answer is: In order to reach the destination we must not exceed the TTL value

5.Daca e sa ne referim la comanda ping, ce afirmație este corectă

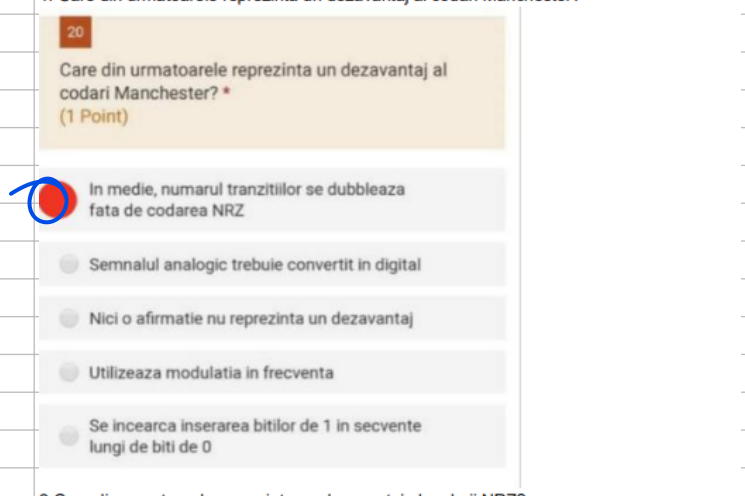
- penultima



se transmit 4 pachete de câte 32 octeți → către o adresă IP

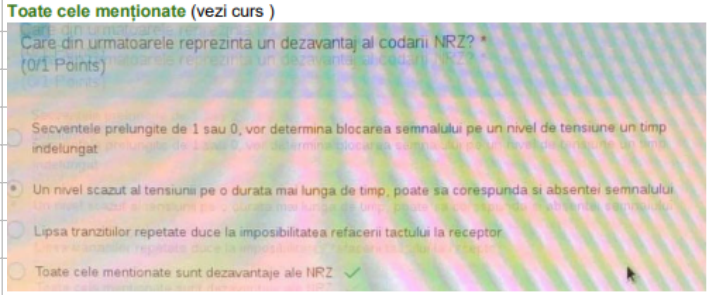
manchester / nrz

1. Care din urmatoarele reprezinta un dezavantaj al codari Manchester?



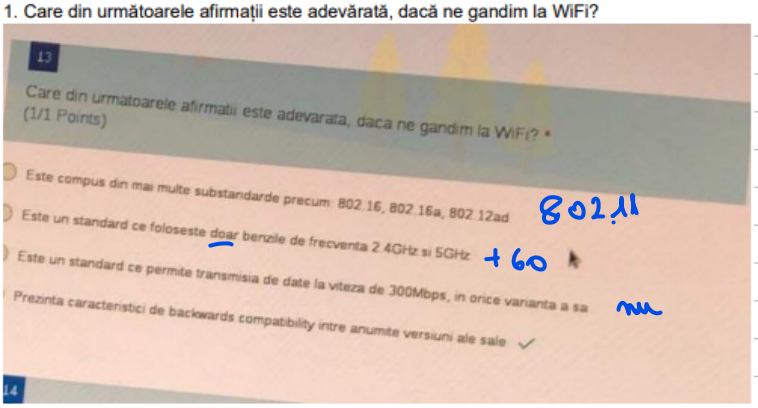
2.Care din urmatoarele reprezinta un dezavantaj al codarii NRZ?

Toate cele menționate (vezi curs)

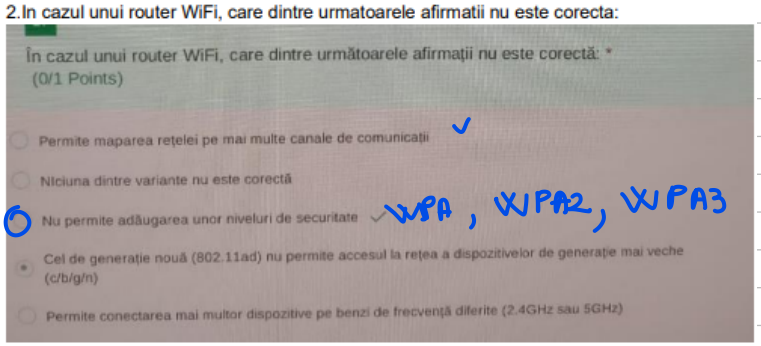


wifi

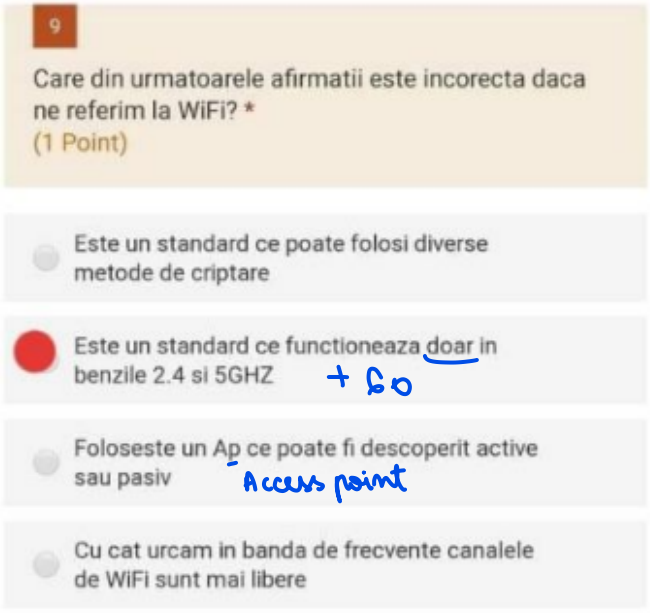
1. Care din următoarele afirmații este adevărată, dacă ne gandim la WiFi?



2.In cazul unui router WiFi, care dintre urmatoarele afirmatii nu este corecta:

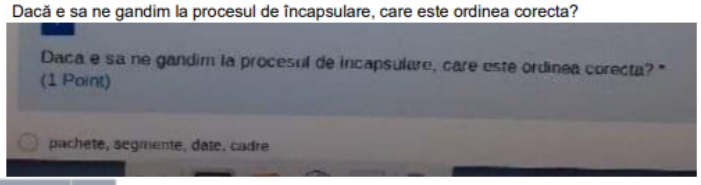


3. Care din următoarele afirmații este incorectă dacă ne referim la wifi?



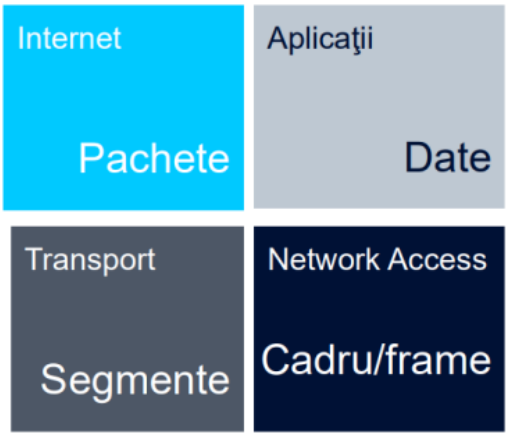
incapsulare

1. Dacă e sa ne gandim la procesul de încapsulare, care este ordinea corecta?



Aplicații	4
Transport	3
Internet	2
Network Access	1

aplicatii transport internet N.A
Date' segmente pachete cadru



icm/ icmp

1.In care din urmatoarele situatii se pot transmite mesaje de eroare, utilizând ICMP-ul?

15

In care din urmatoarele situatii se pot transmite mesaje de eroare, utilizand ICMP-ul? *
(1 Point)

- procesul de reasamblare nu s-a putut efectua
- TTL a atins valoarea zero
- Toate variantele sunt corecte
- checksum-ul header-ului a dat o valoare greşită
- destinatia nu a fost găsită

2. În care din următoarele situații se pot transmite mesaje de eroare, utilizând ICM-ul?

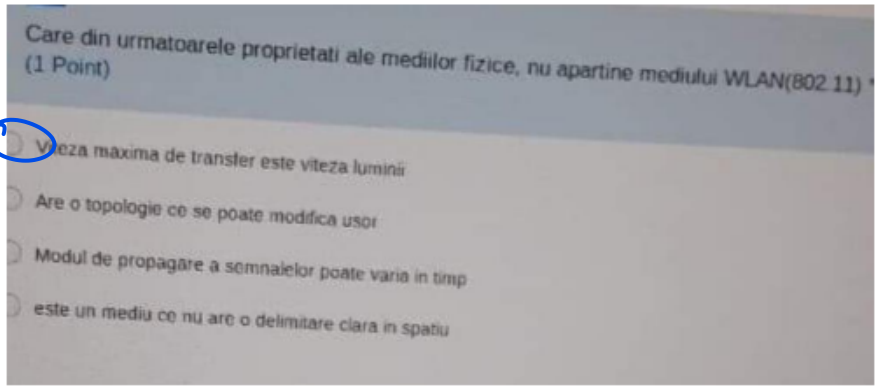
15

In care din urmatoarele situatii se pot transmite mesaje de eroare, utilizand ICMP-ul? *
(1 Point)

- procesul de reasamblare nu s-a putut efectua
- TTL a atins valoarea zero
- Toate variantele sunt corecte
- checksum-ul header-ului a dat o valoare greşită
- destinatia nu a fost găsită

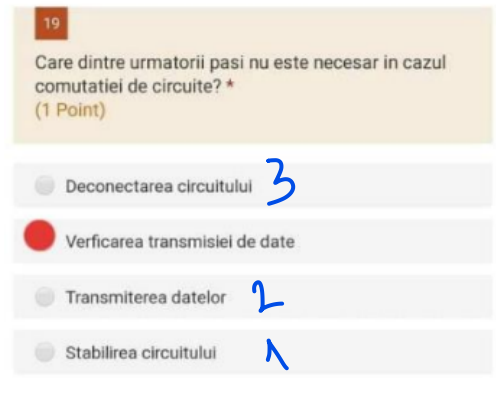
wlan

1.
Care din următoarele proprietăți ale mediilor fizice, nu aparține mediului WLAN?
Viteza maximă de transfer este viteza luminii

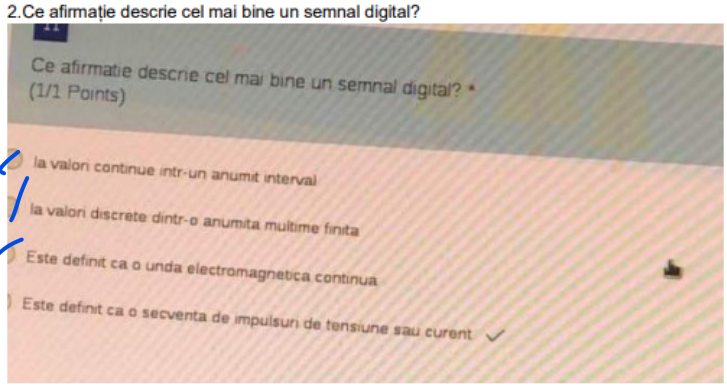


COMUTATIE CIRCUITE/ SEMNAL DIGITAL

1.Care dintre urmatorii pasi nu este necesar in cazul comutației de circuite?

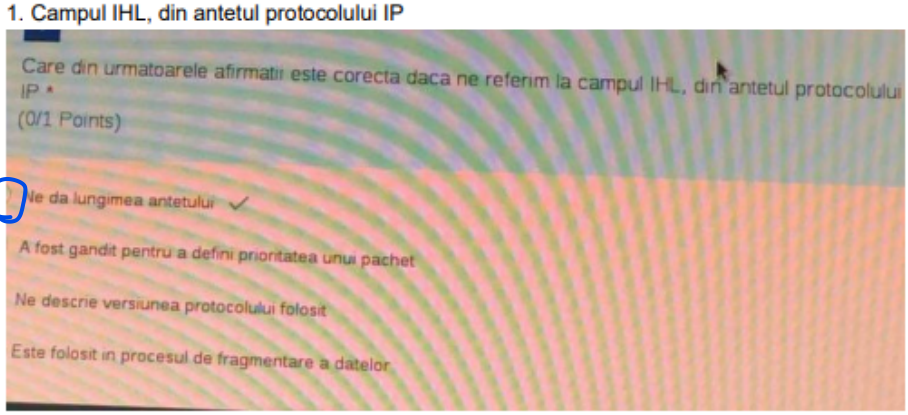


2.Ce afirmație descrie cel mai bine un semnal digital?



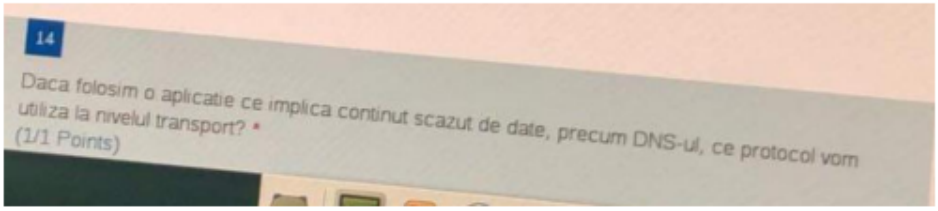
PROTOCOLUL IP

1. Campul IHL, din antetul protocolului IP



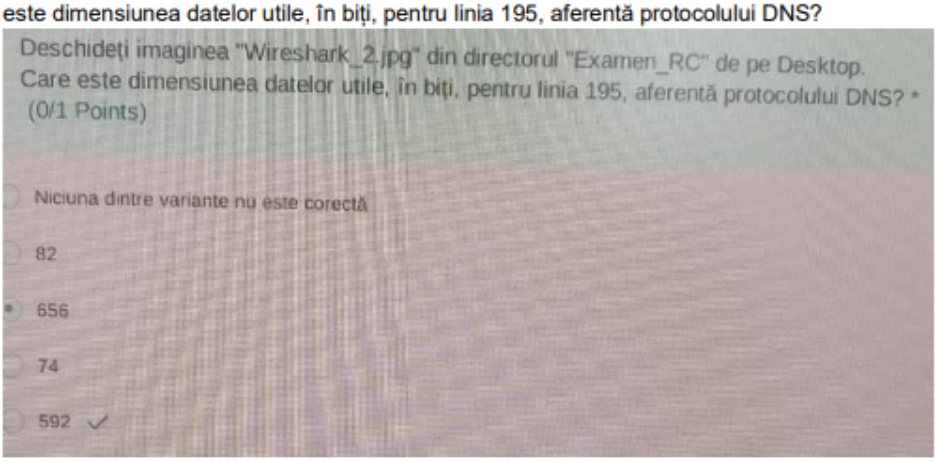
DNS

1.



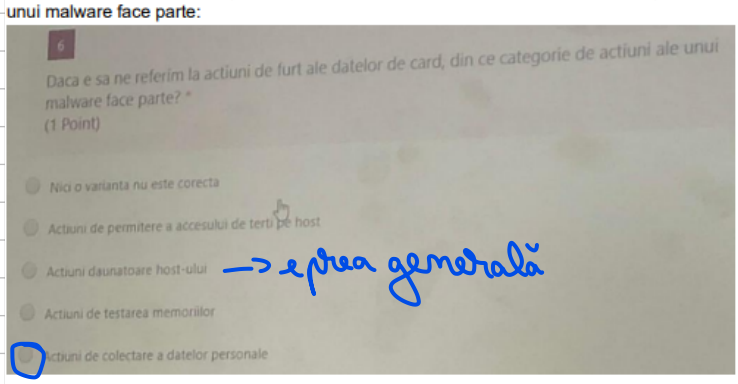
UDP

2.Deschideți imaginea “Wireshark_2.jpg” din directorul “Examen_RC” de pe Desktop. Care este dimensiunea datelor utile, în biți, pentru linia 195, aferentă protocolului DNS?



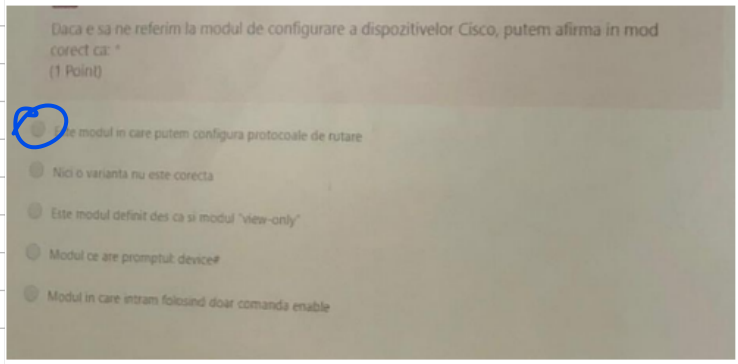
INTREBARI FARA RASPUNS

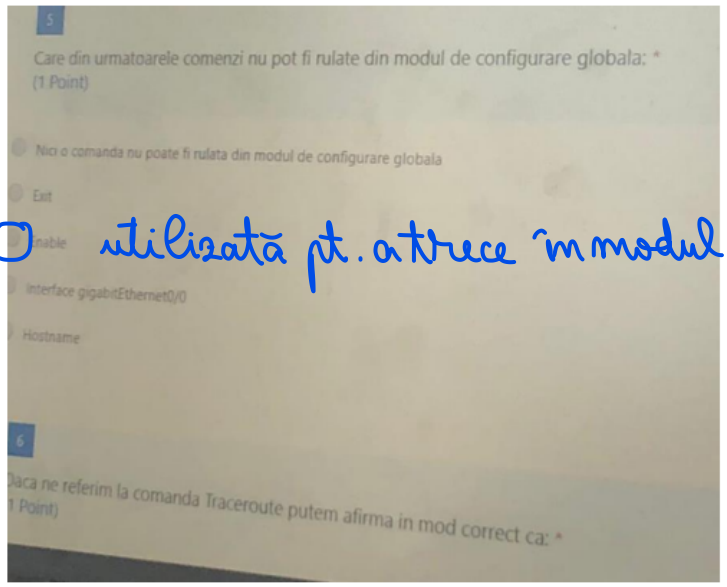
1. Dacă e să ne referim a acțiuni de furt ale datelor de card, din ce categorie de acțiuni ale unui malware face parte:



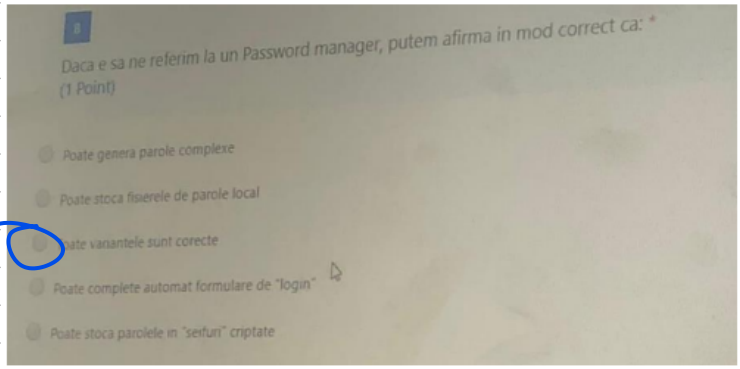
→ e prea generală

2.Dacă e să ne referim la modul de configurare a dispozitivelor Cisco, putem afirma în mod corect că:

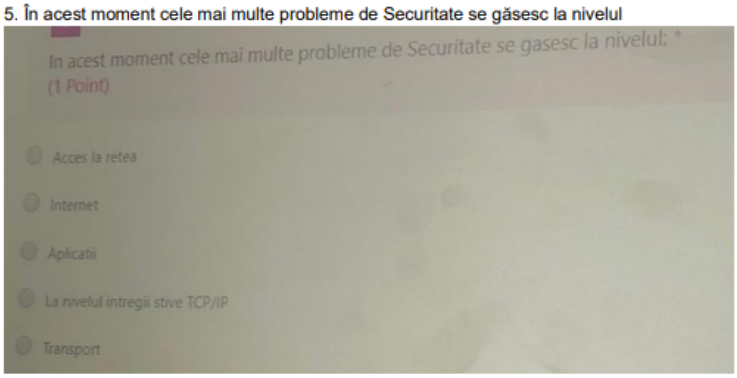




utilizată pt. a trece în modul privilegiat



5. În acest moment cele mai multe probleme de Securitate se găsesc la nivelul



Acces la retea

Problemele de securitate pot aprea la acest nivel, îns acestea sunt de obicei legate de acces neautorizat fizic sau logic la echipamente.
Internet

La acest nivel apar ameninri cibernetice globale, dar nu este neaprat cel mai vulnerabil în momentul actual.
Aplicaii

Acesta este rspunsul corect. Majoritatea problemelor de securitate în prezent provin din aplicaii, datorit vulnerabilitilor de software, atacurilor de tip malware, phishing, sau exploatrii serviciilor online.
La nivelul întregii stive TCP/IP

Este o afirmaie generala si nu specifica un punct concret de vulnerabilitate.
Transport

Problemele la acest nivel sunt mai putin frecvente în comparatie cu cele de la nivelul aplicatiilor.